Team -Code Max
Case Investigation & Problem-Solution Brief

PROBLEM SPACE :- HEALTH CARE

1. Selected Theme & Case Overview

Theme: 03 — Healthcare ("The visit ends, but continuity of care does not.")

Operational Context: A district public hospital manages ~900 outpatient (OPD) visits daily. While clinicians accurately identify patients requiring follow-up, 28% fail to return within the recommended window. The hospital seeks to reduce this loss-to-follow-up (LTFU) via workflow and communication improvements without introducing autonomous clinical diagnosis, treatment modification, or automated emergency triage.

2. Stakeholder Ecosystem & Bottleneck Mapping

Stakeholder Group	Role in Continuity	Core Pain Point / Constraint	Primary Need
Patients & Caregivers	Receive OPD care and execute follow-up visits.	Hard to interpret free-text notes; 19% share household phones; 7% are unreachable; long travel.	Clear, jargon-free clarity on when/why to return and what to bring, via a consented channel.
Doctors & Nurses	Identify follow-up need and author clinical instructions.	Extreme time pressure (~900 visits/day); manual structured notes slow down consultations.	Zero-friction structuring taking <5 seconds to confirm while preserving clinical authority.
Registration & Records	Capture demographics and manage missed-visit logs.	Lack staff capacity to manually call or prioritize hundreds	Fast intake flags for shared/unreachable phones and automated

		of missed follow-ups daily.	administrative queues.
Community Health Workers (CHWs)	Last-mile bridge to rural/peri-urban households.	Lack visibility into which discharged patients have unreachable phones or missed windows.	A clean, consented, village-sorted handoff list with exact follow-up dates and checklists.
Hospital Admin	Oversees OPD quality, safety, and privacy.	Rejects unsafe 'black-box' AI triage; requires strict auditability and privacy compliance.	Safe reduction in the 28% LTFU rate with full human-in-the-loop governance.

3. Evidence, Assumptions & Unknowns Audit

A. CASE EVIDENCE

- Volume: ~900 outpatient visits per day.

- Return Rate: 28% loss-to-follow-up rate.

- Contact Reliability: 19% share a household phone, 7% are unreachable (26% lack direct contact).

- Clinical Risk Boundary: Strict rejection of systems that autonomously diagnose, alter treatment, or triage medical urgency.

B. EXTERNAL EVIDENCE

- Shared Devices: NFHS data shows ~36-42% of rural Indian women rely on shared household devices.

- Driver of LTFU: Research identifies low health literacy and instruction ambiguity as primary non-financial drivers of dropout.

- Regulatory Mandate: DPDP Act 2023 requires explicit, purpose-specific patient consent for data sharing.

C. ASSUMPTIONS (System Design)

- Clinicians will accept an AI structuring step only if review/confirmation takes <5 seconds.

- Capturing the phone owner's relationship and preferred contact window will increase message relay rates.

D. UNKNOWNS

- Exact percentage of the 900 daily visits requiring active follow-up.

- Clinical vs. Administrative split of the 28% non-returning patients.

4. Alternative Problem Framings Considered

Framing Option	Primary Stakeholder	Decision & Rationale
A: Clinical Risk Triage Engine (AI reads charts to rank medical urgency)	Records Staff & Doctors	REJECTED. Violates the core safety boundary. Guessing clinical urgency from sparse notes risks unsafe automated triage.
B: Automated Bulk Reminder Blaster (SMS to all registered numbers)	Hospital IT / Admin	REJECTED. Fails on the 26% lacking personal phones (shared/unreachable), risking privacy breaches on shared devices.
C: End-to-End Instruction & Consented Reachability Chain (SELECTED)	Patients, Doctors, Staff & CHWs	SELECTED. Resolves unstructured notes, 19% shared phones, and 7% unreachable numbers while keeping 100% of clinical decisions human-authoritative.

5. Proposed Solution: CareLoop (4-Link Continuity Chain)

CareLoop addresses the systemic failure through four sequential workflow modules:

Module C1 — Follow-Up Structurer (Consultation Desk):

Parses the doctor's free-text OPD note using a constrained LLM to generate a bilingual Follow-Up Action Card (When, Why, What to Bring). Includes a human-assigned Priority Tag. The card is only saved after the clinician clicks 'Confirm & Sign'.

Module C2 — Shared-Phone Contact Resolver (Registration Desk):

Staff verify phone status (Personal, Shared, Unreachable). For shared phones, the system captures the Phone Holder Relationship and Best Contact Window.

Module C3 — Contact Consent Ledger (Registration Desk):

Creates a timestamped consent record allowing the hospital to share follow-up details with the named alternate caregiver or route to a local CHW if unreachable.

Module C4 — CHW Handoff List (Records & Outreach Desk):

Automatically routes the 7% unreachable patients (and Priority missed patients) into a village-clustered home-visit queue for Community Health Workers, displaying the C1 card details.

6. Synergy Decision Trail (Human-AI Collaboration)

Decision ID & Topic	Initial View & AI Input	Evidence & Verification	Final Human Decision
D1: Problem Framing Selection	Initial: Build AI risk-triage. AI warned of high false-negative risk from sparse text.	Case constraint: Clinical teams reject autonomous diagnosis/triage.	REJECTED. Shifted focus to resolving instruction ambiguity and contact breakdowns.
D2: Handling Shared Phones (19%)	Initial: Bulk automated reminders. AI surfaced NFHS data on shared household device realities.	19% shared phones + DPDP Act privacy rules.	MODIFIED. Added C2 (Shared-Phone Resolver) and C3 (Consent Ledger) modules.
D3: Free-Text Note Parsing (C1)	Initial: LLM texts patient directly. AI noted hallucination	Case boundary: Human clinical	MODIFIED. Enforced strict Doctor

	risk with medical abbreviations.	review must remain authoritative.	Confirmation Gate (1-click verify before saving).
D4: Scope for Unreachable Patients	Initial: Build 4 mobile apps. AI warned of scope creep and suggested a vertical slice.	Hackathon timeframe and evaluation criteria constraints.	ACCEPTED. Scoped build to a unified MERN web app with read-only CHW export list.